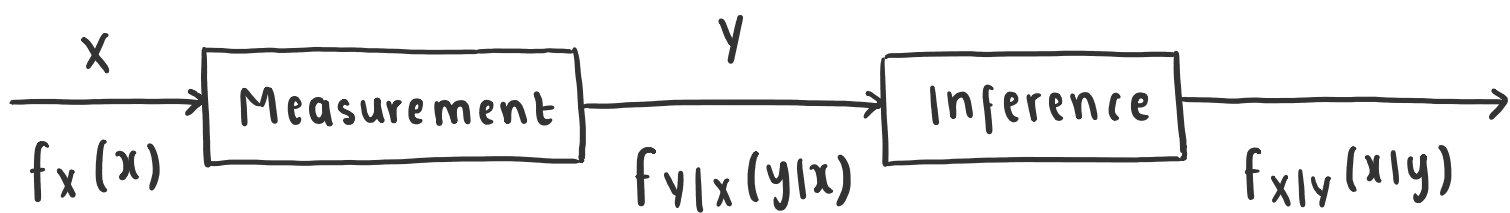


- Inference problem



X : unobservable but known distribution
 We observe measurements of Y given $X=x$.
 OBJECTIVE: draw inference about X having seen Y .

Obtain $f_{X|Y}(x|y)$ using $f_X(x)$ and $f_{Y|X}(y|x)$

- Use Baye's rule :

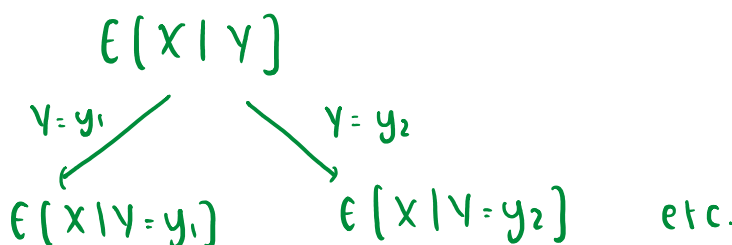
$$f_{X|Y}(x|y) = \frac{f_X(x) \cdot f_{Y|X}(y|x)}{f_Y(y)} = \frac{f_X(x) \cdot f_{Y|X}(y|x)}{\int_{-\infty}^{\infty} f_X(t) \cdot f_{Y|X}(t) dt}$$

Example 3.19, 3.20 (Bertsekas)

- Law of Iterated Expectation

$$E[X] = E\left[E(X|Y)\right]$$

a random variable



$$E(X|Y=y_i) = \sum_x x \cdot p_{X|Y=y_i}(x|y_i)$$

$$E[X] = E(E(X|Y)) = \sum_y E(X|Y=y) \cdot p_Y(y)$$

$$\therefore E[X] = E_Y[E_X(X|Y)]$$

$$E[X \cdot g(Y)|Y] = E[X|Y] \cdot g(Y)$$

pull through property

$$E[X \cdot g(Y) | Y=y_i] = E(X|Y=y_i) \cdot g(y_i)$$

$$E[h(X) \cdot g(Y) | Y] = E[h(X)|Y] \cdot g(Y)$$

- Properties of covariance :

- * $Cov(Z,Z) = Var(Z)$
 - * If X and Y are independent : $Cov(X,Y) = 0$
 - * $Cov(X,Y) = Cov(Y,X)$
 - * $Cov(aX,Y) = a \cdot Cov(X,Y)$
 - * $Cov(X+a,Y) = Cov(X,Y)$
 - * $Cov(X+Z,Y) = Cov(X,Y) + Cov(Z,Y)$
- $\left. \vphantom{\begin{matrix} * \\ * \\ * \\ * \\ * \end{matrix}} \right\} \text{ proved using linearity of expectation}$

$$Cov\left(\sum_{i=1}^m a_i X_i, \sum_{j=1}^n b_j Y_j\right) = \sum_{i=1}^m \sum_{j=1}^n a_i b_j Cov(X_i, Y_j)$$

$$\sum_{i=1}^m a_i X_i$$

- Variance of sum of random variables

$$Z = \sum_{i=1}^n a_i X_i \qquad X_i \rightarrow \text{possibly dependent and non-identical}$$

$$\begin{aligned} Var(Z) &= Cov(Z,Z) \\ &= Cov\left(\sum_{i=1}^n a_i X_i, \sum_{i=1}^n a_i X_i\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j Cov(X_i, X_j) \\ &= \sum_{i=1}^n a_i^2 \cdot Var(X_i) + \sum_{(i,j): i \neq j} a_i a_j Cov(X_i, X_j) \end{aligned}$$

- * If X_i 's are independent , $Cov(X_i, X_j)=0$

$$\Rightarrow Var(Z) = \sum_{i=1}^n a_i^2 Var(X_i)$$

Q. Consider $\{X_i, i=1,2,\dots,n\}$. All X_i are iid.

Consider $S_n = \frac{\sum_{i=1}^n X_i}{n}$.

Show that $Var(S_n) = \frac{Var(X)}{n}$

ans $S_n = \sum_{i=1}^n X_i$

$$\begin{aligned} Var(S_n) &= Cov\left(\frac{\sum_{i=1}^n X_i}{n}, \frac{\sum_{i=1}^n X_i}{n}\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n Cov(X_i, X_j) = \frac{1}{n^2} \sum_{i=1}^n Var(X_i) \\ &= \frac{n \cdot Var(X)}{n^2} = \frac{Var(X)}{n} \end{aligned}$$